DETSI Wide Bay Report

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1 Executive Summary

2 Background

Accurately characterizing drivers and resilience of seagrass cover is essential to improving the ecological health of the Wide Bay ecosystem. Over the past few decades, various scientific efforts have informed our current understanding of trends, behaviors, and flood responses of seagrass cover in Wide Bay [Preen et al., 1995, McMahon et al., 2005, York et al., 2022] as well as Australia more widely [Kilminster et al., 2015, Gilby et al., 2018, O'Brien et al., 2018]. Remote sensing has offered new capabilities for seagrass cover mapping [Roelfsema and Phinn, 2009], while innovative statistical approaches have offered novel methods for studying resilience at site-specific scales [Gilby et al., 2023]. Our work contributes to this body of literature by applying spatially explicit statistical models [Cressie, 1993] to study seagrass cover drivers, resilience, and recovery in Wide Bay. Specifically, we provide and contextualize answers to four specific research questions, applicable to Wide Bay:

1. Can we identify water quality covariates that act as triggers for seagrass cover management responses? Are these triggers different based on whether they result from a specific event (e.g., flood) or are ambient?
2. Did seagrass cover improve after the flood events? If so, how and where?
3. Can we identify site-specific resilience indicators, which describe where recovery is likely to be enhanced or impeded?
4. What factors contribute to seagrass resilience?

The spatially explicit statistical models (i.e., spatial models) we used to answer these research questions are related to commonly used linear and generalized linear models but incorporate *spatial dependence*. Spatial dependence measures how similar two sites are based on their distance. Formally, we can build spatial dependence directly into spatial models by leveraging Tobler's First Law of Geography [Tobler, 1970], which states succinctly that "nearby sites tend to be more similar than distant sites". Not only is this intuitive based on "first principles" perspective, but spatial models to provide many advantages for modeling spatial data compared to models that assume independence among observations (i.e., nonspatial models). First, nonspatial models tend to underestimate uncertainty, yielding covariate standard errors and p-values for covariates that are too small and corresponding confidence intervals that are too narrow. Second, spatial models improve prediction at unsampled locations by borrowing strength from nearby observations, yielding more precise predictions and corresponding standard errors that vary by location. Third, spatial models decompose model error into separate spatial and nonspatial components, informing residual and resilience structure in ways that nonspatial models cannot. Fourth, spatial models can provide ecologically meaningful insight into the spatial dependence structure of the process, informing relationships among measured and unmeasured variables as well as future sampling plans. Zimmerman and Ver Hoef [2024] provide a thorough review of spatial models for ecological and environmental data, providing more detail and justification for the claims made above.

Spatial models offer many benefits over commonly used machine learning algorithms like random forests [Breiman, 2001, Cutler et al., 2007, James et al., 2017, Kuhn and Silge, 2022]. First, traditional machine learning algorithms do not incorporate spatial dependence and, by consequence, often perform worse than spatial models at characterizing important covariates and predicting at unobserved locations [Fox et al., 2020]. Second, machine learning algorithms struggle to quantify uncertainty in a principled manner, something spatial models excel at.

Third, covariate inference is more straightforward using spatial models, as they naturally provide p-values that enable structured testing of hypotheses. Fourth, spatial models provide a rich set of diagnostics like leverage and Cook's distances [Montgomery et al., 2021], quantities not generally defined for machine learning algorithms. This is not to say that machine learning algorithms cannot be very useful in ecological data analyses, but rather than spatial models are sometimes a more effective tool for modeling spatial data.

Though there are many benefits of spatial models over machine learning algorithms, the reverse is also true. Machine learning algorithms much more simply capture complex nonlinearities "out-of-the-box", bypassing the tuning often required for spatial models that involve nonlinear components like splines. This can be especially useful for ecological data, which are often full of nonlinear relationships [Fox et al., 2017]. Machine learning algorithms also typically rely on few, if any, distributional assumptions and can be more robust against unusual observations like outliers [Hastie et al., 2009]. Machine learning algorithms also tend to be much more computational efficient than spatial models. Importantly, machine learning algorithms and spatial models can be complementary tools that help shed insight into complicated ecological processes (something we reiterate in the upcoming analyses). An exciting and developing area of statistical research involves incorporating spatial modeling strategies directly into machine learning algorithms [Saha et al., 2023, Heaton et al., 2025].

The first spatially explicit model we define is the spatial linear model. The spatial linear model is a *mixed-effects model* [Pinheiro and Bates, 2000, Bolker, 2015] written as

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + ... + \beta_p x_{p,i} + \tau_i + \epsilon_i, \tag{1}$$

where $i = 1, 2, ..., n$ indexes the n observations. The quantity y_i is the response variable (for observation i), the β parameters are slope parameters controlling the covariates, $x_{j,i}$ (for $j = 1, 2, ..., p$). The random error τ_i is a spatial random error, and the random error ϵ_i is the nonspatial (i.e., independent) random error. The spatial random error is spatially structured, capturing spatial residual effects like environmental gradients we do not have data to fully represent. The nonspatial random error is not spatially structure and captures nonspatial residual effects like measurement error and microscale variation. The difference between a traditional (i.e., nonspatial) linear model and a spatial linear model is the inclusion of τ_i (in the spatial linear model). The spatial random error implied by τ_i is governed by a *spatial covariance function*. The spatial covariance function allows the model to incorporate spatial dependence, yielding the improved model performance previously described. Figure 1 shows three different spatial covariance functions that describe different ways the spatial covariance decays with distance. Model parameters and standard errors are estimated from data using likelihood-based [Patterson and Thompson, 1971, Harville, 1977, Wolfinger et al., 1994] or semivariogram-based [Cressie, 1985, Curriero and Lele, 1999] estimation methods. Often, the spatial linear model in Equation 1 in is written more compactly using *matrix notation*:

$$\mathbf{y} = \mathbf{X}\beta + \tau + \epsilon, \tag{2}$$

where $\mathbf{X}$ is a matrix that holds the j covariates for each of n observations, and $\mathbf{y}$, β, τ, and τ are vectors that contain each of the n or j quantities from Equation 1. The spatial linear model can be adjusted to account for nonspatial random effects just as one would add random effects to a nonspatial linear mixed model:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{u} + \tau + \epsilon, \tag{3}$$

where $\mathbf{Z}$ is a random effect design matrix (dimension $n \times k$) that indexes each of the k random effects in $\mathbf{u}$. For more information about specific forms of various spatial covariance functions, see Schabenberger and Gotway [2017] and Zimmerman and Ver Hoef [2024].

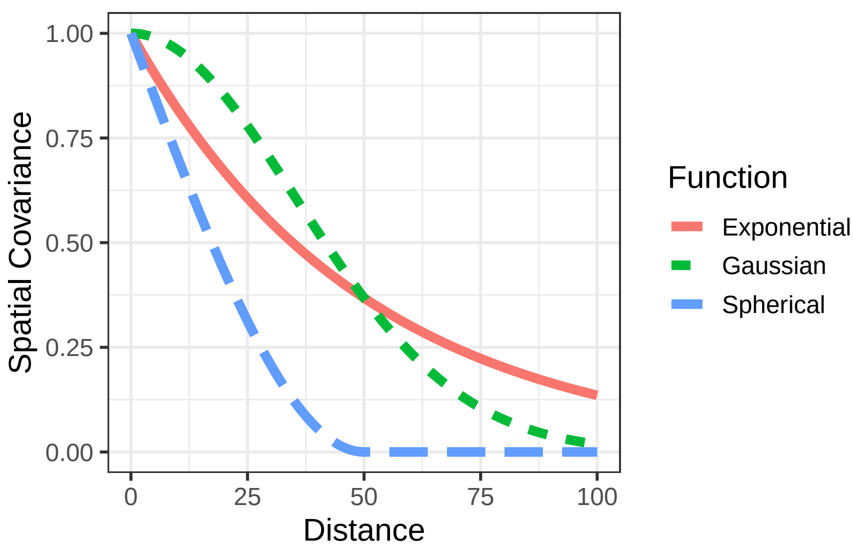

Figure 1: Three spatial covariance functions that represent different spatial dependence behavior.

Like the traditional linear model, the spatial linear model can capture complex nonlinearities in the data through the use of interaction effects [Faraway, 2002, Lenth and Piaskowski, 2025], splines [Chambers and Hastie, 1992, Ruppert et al., 2003], additive models [Wood, 2017, Pedersen et al., 2019], and penalized regression models like the ridge and lasso [Tibshirani, 1996]. Generally, this is accomplished by leveraging *basis functions* that break down nonlinear behavior into a set of additive linear components that may include penalty terms which guard against overfitting. Furthermore, there is no assumption that y_i itself is normally distributed; rather, distributional assumptions are placed on the random errors after incorporating the slope parameters. This clarification implies the spatial (and nonspatial) linear models can be quite effective at modeling a range of data that are not normally distributed. For highly nonnormal data, tools like generalized linear models, which we discuss later, can be helpful. Finally, under general conditions, the slope parameters of the spatial (and nonspatial) linear models are approximately normally distributed with a large sample size (see Casella and Berger [2024] for a general justification via the Central Limit Theorem and Zhang and Zimmerman [2005] for nuances regarding spatial data). This is beneficial because it justifies valid hypothesis testing on the slope coefficients (often of primary ecological interest), no matter the distribution of the data (given certain assumptions are satisfied).

One of the most useful applications of the spatial linear model is predicting the response variable at unobserved locations. A very useful spatial prediction method called Kriging has a rich theoretical justification [Cressie, 1990, Stein, 1999]. Kriging is the spatial analogue to best linear unbiased prediction (BLUP) for mixed models [Henderson, 1975]. Via Kriging, the spatial linear model produces BLUPs at each unobserved location of interest, alongside prediction intervals to characterize uncertainty. While useful, sometimes the goal is to create a BLUP for an entire region of interest. This technique, which builds upon the theoretical foundations of Kriging, is called block Kriging [Ver Hoef, 2002, Cressie, 2006]. The intuition is to make point predictions at a very dense grid in the study region and then average these predictions, appropriately characterizing uncertainty. Together, Kriging and block Kriging provide a complete set of tools for spatial prediction, given the fit of a spatial linear model.

Spatial generalized linear models are an extension of spatial linear models to highly nonnormal data [Christensen and Waagepetersen, 2002, Zhang, 2002, Diggle and Ribeiro Jr, 2007, Bonat and Ribeiro Jr, 2016]. Ver Hoef et al. [2024] proposed a novel application of the Laplace approximation to enable restricted maximum likelihood estimation of spatial generalized linear models. Their model formulation builds upon Equation 1 and is given by

$$g(\mu_i) = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + ... + \beta_p x_{p,i} + \tau_i + \epsilon_i, \tag{4}$$

where $g(\mu_i)$ is a "link function" that links the distribution of y_i to a linear function of its mean, μ_i. Common link functions include the log link for count and skewed data and the logit link for binomial and proportion data (Table 1). Using the same argument as for spatial linear models, nonspatial random effects can be added to spatial generalized linear models.

Table 1: Common generalized linear model families, link functions, and appropriate data types.

Family	Link Function	Link Name	Data Type
Binomial	$f(\mu) = \log(\mu/(1-\mu))$	Logit	Binary; Binary Count
Poisson	$f(\mu) = \log(\mu)$	Log	Count
Negative Binomial	$f(\mu) = \log(\mu)$	Log	Count
Beta	$f(\mu) = \log(\mu/(1-\mu))$	Logit	Proportion
Gamma	$f(\mu) = \log(\mu)$	Log	Skewed
Inverse Gaussian	$f(\mu) = \log(\mu)$	Log	Skewed

Spatial models are complicated, both theoretically and computationally. This has contributed to their relative lack of use in the ecological community. However, recent advances in open-source software have made spatial models more accessible to ecologists who have strong subject matter expertise but may lack sufficient statistical background or computational skills to implement them from scratch. One such advancement is the `spmodel` **R** package [Dumelle et al., 2023], which emulates base-**R** functions like `lm()` and `glm()` to account for spatial dependence via `splm()` and `spglm()`, respectively. Familiar **R** functions like `summary()` and `predict()` can be used directly on models fit via `spmodel`. Importantly, `spmodel` also has tools for spatial random forest residual modeling [Fox et al., 2020], applications to large data of several thousand observations via spatial indexing [Ver Hoef et al., 2023], and cross validation [Stone, 1974, Roberts et al., 2017].

3 The Data

We started with nearly 4,000 seagrass cover observations in Wide Bay across three financial years (2021-2022, 2022-2023, and 2023-2024). Some data (approximately 1,000 observations) were available from 1998-2002 but were omitted as they were far away temporally and did not have corresponding eReefs data [Steven et al., 2019]. In total, eReefs data with modeled water quality covariates related to turbidity and salinity were available at 2,663 observations. Because one of our research questions involved connecting water quality covariates to management actions, we proceed with this subset of 2,663 observations for modeling. Of the 2,663 observations, 1,073 were in 2021-2022, 524 were in 2022-2023, and 2,663 were in 2023-2024 (Table 2).

Table 2: Total seagrass observations in Wide Bay by financial year.

Financial Year	Sample Size	Percent (of Total)
2021-2022	1,073	40.3%
2022-2023	524	19.7%
2023-2024	1,066	40.0%
Total	2,663	100.0%

4 Water Quality and Geographic Modeling

Total cover is a percentage that ranges from 0% (no seagrass) to 68.3%. Figure 2 shows total cover by financial year throughout the Bay. Total cover tends to be largest along the eastern shoreline of the Bay and lowest in the middle of the Bay. Figure 2 also reveals that 2022-2023 lacked sampling in much of the Great Sandy Strait.

We considered for modeling several water quality and geographic covariates. Table 3 summarizes the water quality variables, which included both ambient and flood-related metrics related to turbidity, salinity, and temperature. The “Variable” column indicates the water quality variable turbidity, salinity, or temperature. The “Flood” variable indicates whether

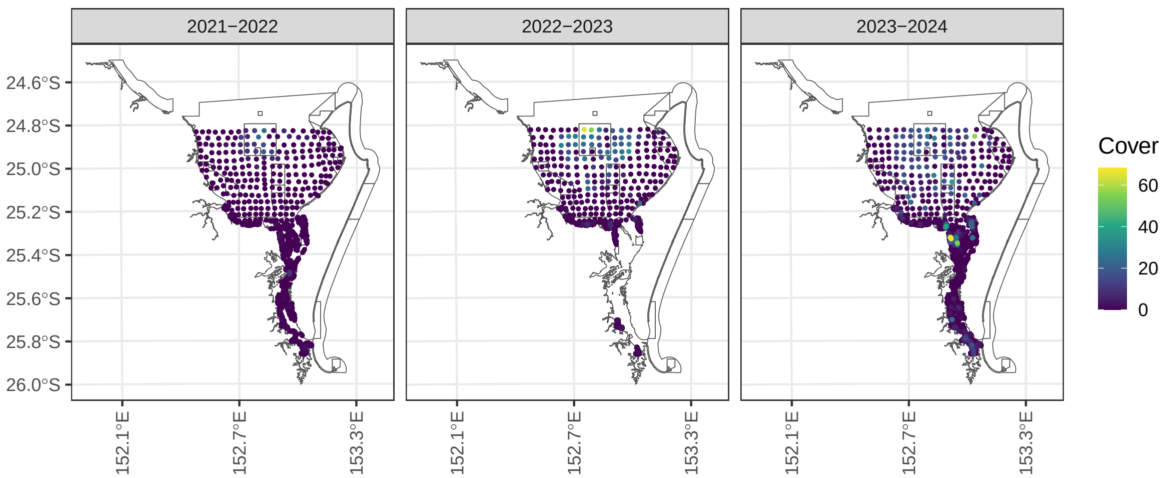

Figure 2: Total seagrass cover in Wide Bay by financial year.

the measurement corresponds to a flood variable or not. The “Period” variable indicates the prior period (in days) relative to sampling or indicates whether the variable corresponds to a flood event. The “Type” column is the type of statistic, either a mean, minimum, maximum, number of days above (or below) a cutoff during “Period”, or a maximum spell (i.e., duration) above (or below) a cutoff during “Period”. The “Included” variable indicates whether the variable was included as a covariate in the final spatial model. We included only a few variables in the final model: mean turbidity in the 90 days prior to sampling, average mean turbidity during both flood events, and mean temperature in the 90 days prior to sampling. We only included a single turbidity metric prior to sampling because these variables were all quite correlated with one another, making it hard for the model to tease apart signals. We believed 90 day mean turbidity prior to sampling to be the most accurate “running average” of turbidity prior to sampling. We used a similar rationale to select only a single turbidity metric from the flood and a single temperature metric prior to sampling. We omitted a flood salinity metric primarily because we lacked data for salinity prior to sampling, thus it would be challenging to establish a “baseline” for salinity in order to tease apart prior to sampling effects from flood effects (moreover, they did not seem strongly related to total cover). We provide more detail regarding the statistical modeling in Section 4.1.

Table 3: Wide Bay water quality variables considered for modeling.

Variable	Flood	Period	Type	Cutoff	Included
Turbidity (NTU)	No	7	Mean	NA	No
Turbidity (NTU)	No	30	Mean	NA	No
Turbidity (NTU)	No	90	Mean	NA	Yes
Turbidity (NTU)	No	7	Above	3	No
Turbidity (NTU)	No	30	Above	3	No
Turbidity (NTU)	No	90	Above	3	No
Turbidity (NTU)	No	7	Above	5	No
Turbidity (NTU)	No	30	Above	5	No
Turbidity (NTU)	No	90	Above	5	No
Turbidity (NTU)	No	7	Above	6	No
Turbidity (NTU)	No	30	Above	6	No
Turbidity (NTU)	No	90	Above	6	No
Turbidity (NTU)	Yes	Flood	Above	3	No
Turbidity (NTU)	Yes	Flood	Above	5	No
Turbidity (NTU)	Yes	Flood	Above	6	No
Turbidity (NTU)	Yes	Flood	Duration	3	No
Turbidity (NTU)	Yes	Flood	Duration	5	No
Turbidity (NTU)	Yes	Flood	Duration	6	No
Turbidity (NTU)	Yes	Flood	Mean	NA	Yes
Turbidity (NTU)	Yes	Flood	Max	NA	No
Salinity (PSU)	No	7	Below	20	No

Table 3: Wide Bay water quality variables considered for modeling.

Variable	Flood	Period	Type	Cutoff	Included
Salinity (PSU)	No	30	Below	20	No
Salinity (PSU)	No	90	Below	20	No
Salinity (PSU)	No	7	Below	25	No
Salinity (PSU)	No	30	Below	25	No
Salinity (PSU)	No	90	Below	25	No
Salinity (PSU)	No	7	Below	30	No
Salinity (PSU)	No	30	Below	30	No
Salinity (PSU)	No	90	Below	30	No
Salinity (PSU)	Yes	Flood	Below	20	No
Salinity (PSU)	Yes	Flood	Below	25	No
Salinity (PSU)	Yes	Flood	Below	30	No
Salinity (PSU)	Yes	Flood	Duration	20	No
Salinity (PSU)	Yes	Flood	Duration	25	No
Salinity (PSU)	Yes	Flood	Duration	30	No
Salinity (PSU)	Yes	Flood	Mean	NA	No
Salinity (PSU)	Yes	Flood	Min	NA	No
Temperature (deg C)	No	7	Mean	NA	No
Temperature (deg C)	No	30	Mean	NA	No
Temperature (deg C)	No	90	Mean	NA	Yes

Table 4 summarizes the geographic variables, which have the following structure:

- The “Sediment Type” variable has five levels: Sand, Sandy Mud, Muddy Sand, Mud, and Rock.
- The “Tidal Range” variable has three levels: Shallow Subtidal, Intertidal, and Deep Subtidal.
- The “Years Since Flood (One)” is numeric from 0.096 (May, 2022) to 1.65 (November, 2023).
- The “Years Since Flood (Two)” is numeric from 0 (Pre-flood) to 1.40 (November, 2023).
- The “Algae Presence” variable has two levels: Algae present and algae not present
- The “Algae Cover” variable is numeric from zero to one.
- The “Growing Season” variable has two levels: Growing (September - December) and Not Growing (January - August).
- The “Zone Class” variable has six levels: Central, East, Great Sandy Strait, Hervey Bay, Southern, and Southwest (Figure 3).

Table 4: Wide Bay geographic variables considered for modeling.

Variable	Effect	Included
Sediment Type	Fixed	Yes
Tidal Range	Fixed	Yes
Years Since Flood One	Fixed	Yes
Years Since Flood Two	Fixed	No
Algae Presence	Fixed	No
Algae Cover	Fixed	No
Growing Season	Fixed	No
Zone Class	Random	Yes

From Table 4, we kept the sediment type, tidal range, and years since flood (one) variables as covariates in the model. We only included time since the first flood, as nearly all the data aside from teh observations in May 2022 were after both floods, making it possible to study cumulative flood effects but harder to tease apart effects from each individual flood. The algae variables were removed, as they did not seem related to total cover. We removed growing season from the model after determining that it likely is not ecologically useful, as a simple month proxy likely does not adequately account for the complexities of the more nuanced aspects of seagrass growth with time. We used zone class as a random effect to account for within-class correlation that may impact the relationships among total cover and the remaining variables. We provide more detail regarding the modeling in Section 4.1.

4.1 Model Fit

Table 5: Variables in the final Wide Bay total cover model, their type (numeric or categorical), and if categorical, the number of unique levels.

Variable	Type	No. Levels
90 Day Temperature	Numeric	NA
90 Day Turbidity	Numeric	NA
Mean Floods Turbidity/Years Since Flood One^2	Numeric	NA
Sediment Type	Categorical	5
Tidal Range	Categorical	3

To describe total seagrass cover in Wide Bay, our final model was a spatial linear model (Equation 1) with the following explanatory variables: tidal range, sediment type, 90 day mean temperature (prior to sampling), 90 day turbidity (prior to sampling), and mean turbidity during the flood periods scaled by time since the first flood squared (Table 5). We scaled

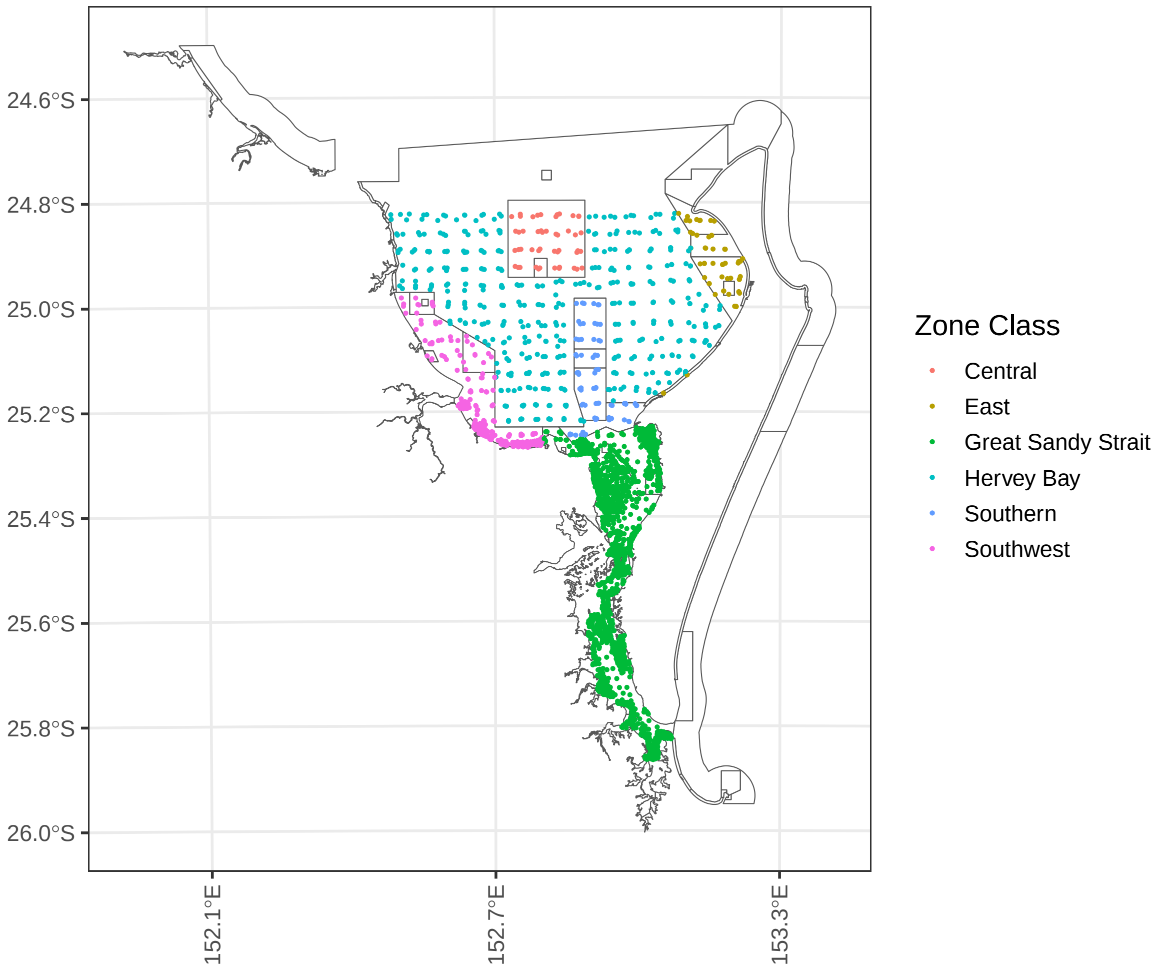

Figure 3: Observations and their zone classes in Wide Bay.

mean turbidity during the first flood period so that the effect of turbidity during the flood event was allowed to decay with time since the flood. Intuitively, this means that with time since the flood, the impact of turbidity during the flood event is lessened. We also considered scaling by time since flood (not squared), which produced similar results. Furthermore, we used an exponential spatial covariance function and included a random effect for zone class. We fit the model using restricted maximum likelihood via the `splm()` function in `spmodel`.

Table 6 shows the results from an analysis of variance applied to the fitted model. The sediment type and temperature covariates had small p-values (p-value < 0.01) indicating significant evidence of a relationship with total cover, while other covariates had large p-values (p-value > 0.1) indicating little evidence of an association with total cover; here are some relevant takeaways:

- Sediment Type: The estimated (marginal) average total cover is largest in mud (4.3%), followed by sand (3.6%), muddy sand (3.2%), rock (3.0%), and sandy mud (2.8%).
- 90 Day Temperature: A one-degree (C) increase in 90 day temperature is associated with a decrease in average total cover by 0.5% points.

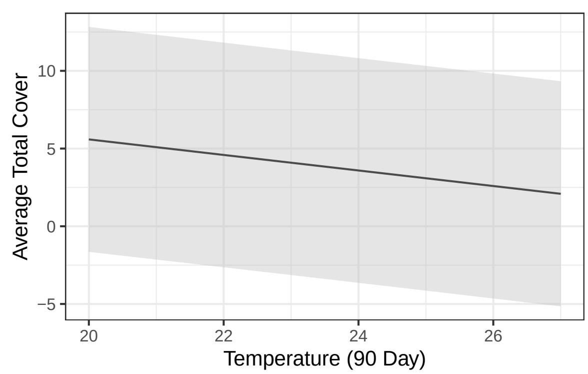

Figure 4: Average total cover as a function of 90 day mean temperature. Estimates (and confidence intervals) were evaluated at the means of all other numeric variables, and for the intertidal range and Sand sediment type (the same trends occur for both variables among the remaining species and subzones).

Together, these results can inform triggers for management responses in various conditions. For example, given the decrease in total cover with temperature, one could propose a specific temperature threshold above which management intervention is warranted. Note that the estimated marginal means for tidal range and sediment type were calculated using the `emmeans` **R** package [Lenth and Piaskowski, 2025].

Table 6: ANOVA table of covariates in the Wide Bay water quality and geographic model.

Effects	df	Chi-sq	p.value
90 Day Temperature	1	54.665	0.000
Sediment Type	4	16.319	0.003
Tidal Range	2	4.106	0.128
90 Day Turbidity	1	1.963	0.161
Mean Flood Turbidity/Years Since Flood One^2	1	0.007	0.934

We may inspect model diagnostics to better understand the overall model fit. Here we discuss how to partition the model into different components of variability, infer decay behavior of spatial autocorrlation, analyze standardized residuals to assess model assumptions, and identify influential observations that have a sizable impact on model fit. From Table 7, we see that of the total model variability, 5.4% is attributable to the explanatory variables (this quantity is sometimes called the Pseudo R-squared), 54.2% is attributable to the spatial random error, 5.8% is attributable to the zone class, and the remaining variability is independent random error. The estimated exponential spatial range (ϕ) is 14,757 meters, which suggests two total cover observations are approximately (spatially) uncorrelated at a distance of $3\phi \approx 45$km, or forty-five kilometers. While the raw residuals will inform our resilience modeling in Section 5, we can check certain model assumptions using the standardized residuals, which have been decorrelated. Generally, the standardized residuals are centered around zero, though there is a slight right skew suggesting some observed total cover values are larger than expected (according to the model). Observations are called "influential" if they highly influence model fit. Influence is often measured via Cook's distance, and no observations exceed the often-used Cook's distance threshold value of one [Cook and Weisberg, 1982].

Table 7: Variance components in the Wide Bay water quality and geographic model.

Variance Component	Percent (of Total Variability)	Parameter Value
Covariates	5.4%	NA
Spatial Effect	63.5%	50.3
Independent Error	23.1%	18.3
Zone Class	8.0%	24324.6

4.2 Block Kriging

A tremendous benefit of using the spatial linear model is that block Kriging can be directly applied to predict the mean total seagrass cover bay-wide for any combination of explanatory variables and geographic regions, even if these variables were not directly used in modeling. Figure 5 and Figure 6 show that generally, there is no evidence total cover in the zone classes or conversations zones is decreasing since the floods (in financial year 2021-2022). While zone class was included in the model as a random effect, conservation zone type was not – via block Kriging, we can still quantify broad trends in total seagrass cover. Figure 7 shows that total seagrass cover is improving in the deep subtidal and intertidal classes since the floods, while Figure 8 shows that total seagrass cover is recovering bay-wide since the floods. Another useful application of block Kriging is scenario modeling, which can be used to characterize the impact of various future changes to broad-scale climatic patterns.

4.3 Alternative Approaches

In addition to the spatial linear model, we considered several different modeling approaches (all using the same explanatory variables): a nonspatial linear model, a spatial beta regression model, a spatial gamma regression model, a random forest model, and a spatial binomial model. The nonspatial linear model provided a useful baseline to compare the spatial linear model against, elucidating the impact of spatial dependence on model fit. The spatial beta regression model is a generalized linear model used to characterize proportion data between zero and one. To adhere to this restriction, total cover was 1) scaled to a proportion and 2) if exactly zero (one), set to 0.01 (0.99). The spatial gamma regression model is a generalized linear model used to characterize skewed positive data. To adhere to this restriction, if total cover was exactly zero, it was set to 0.01. The random forest provided a useful comparison to a machine learning approach, quite different than the aforementioned statistical approaches. We characterized the directionality of random forest effects using partial dependence plots (Figure 9). Finally, the binomial model is a generalized linear model used to characterize presence/absence (success/failure) data. To adhere to this restriction, total cover was 1) given a value of zero if there was no seagrass or 2) given a value of one if there was any seagrass.

All six approaches found notable, negative relationships between temperature and total cover as well as notable to moderate relationships between sediment type and total cover as well as tidal range and total cover. The nonspatial linear model, random forest, and spatial beta regression model found notable, generally negative relationships between turbidity and total cover, although quite small in magnitude. From Table 8, the spatial linear model and random forest had the lowest out-of-sample root-mean-squared-prediction error, followed by the generalized linear models and then by the nonspatial linear model. Random forest partial dependence plots [Greenwell, 2017], which characterize the marginal behavior of a variable in a random forest model, suggested linear effects of the model variables to be reasonable. Given this context, we chose the spatial linear model as the final model, although it is certainly reasonable to justify other modeling choices that enforce specific, more realistic restrictions on the response distribution like the spatial beta regression model (enforcing proportions).

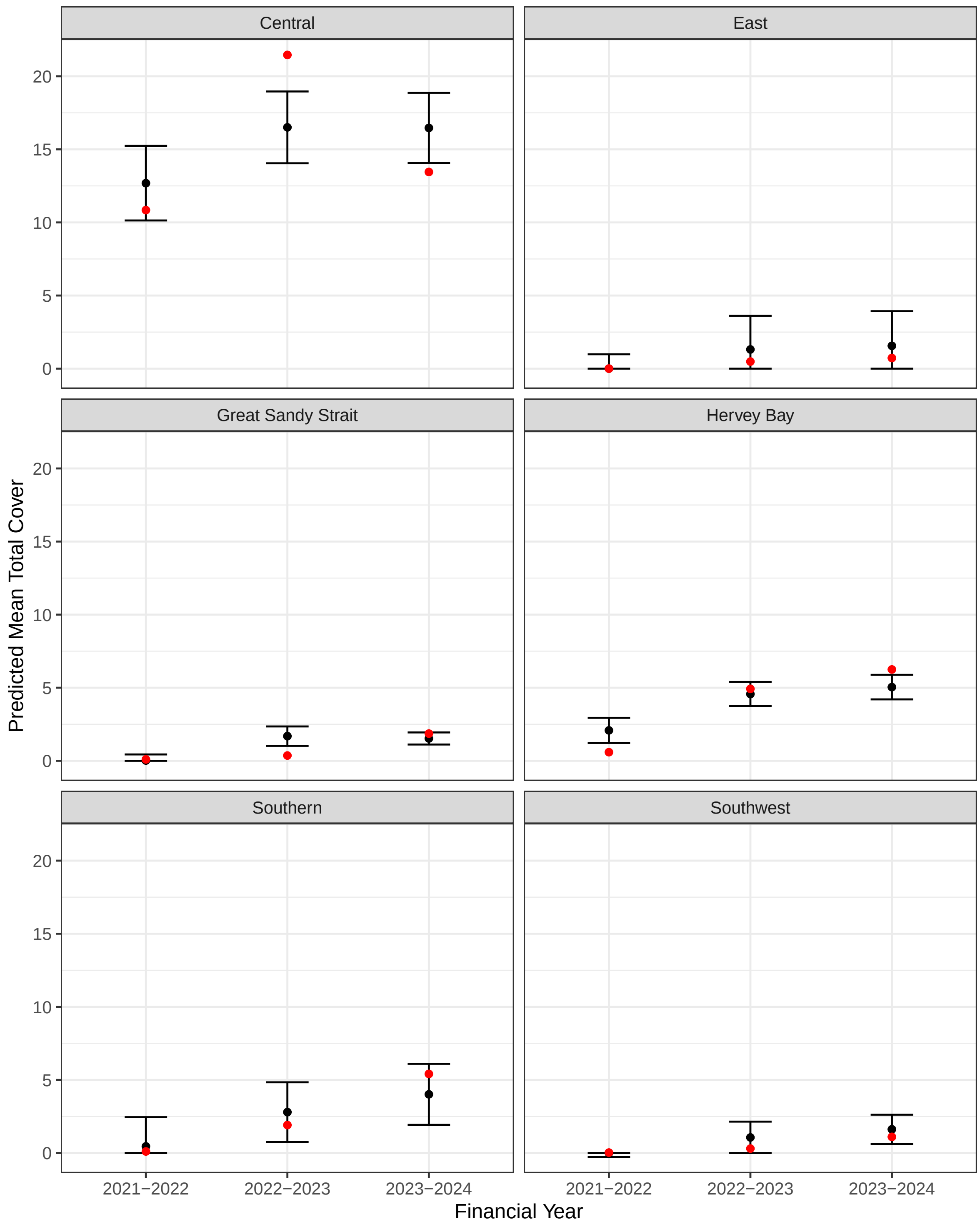

Figure 5: Average total cover predictions by zone class. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

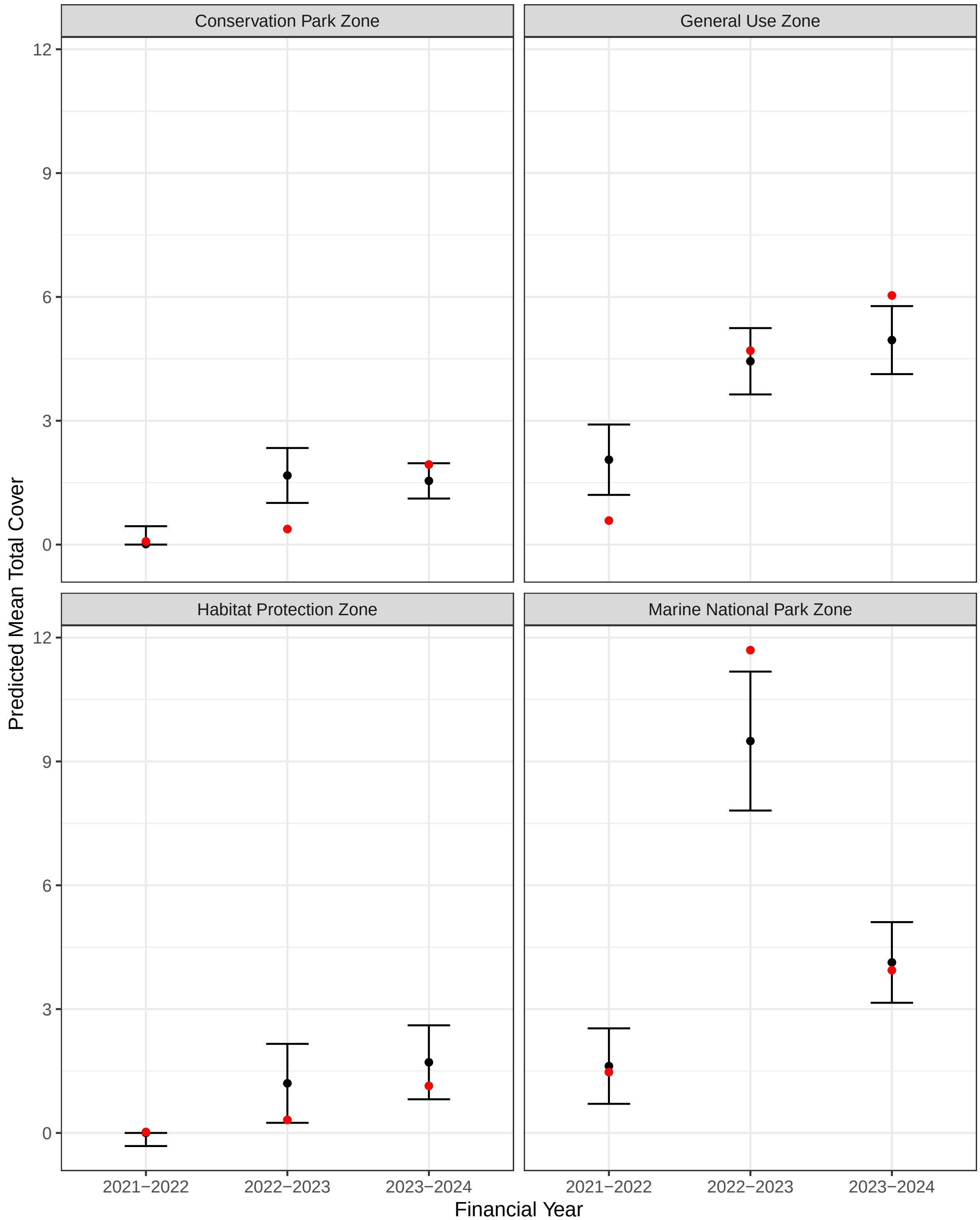

Figure 6: Average total cover predictions by conservation zone. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

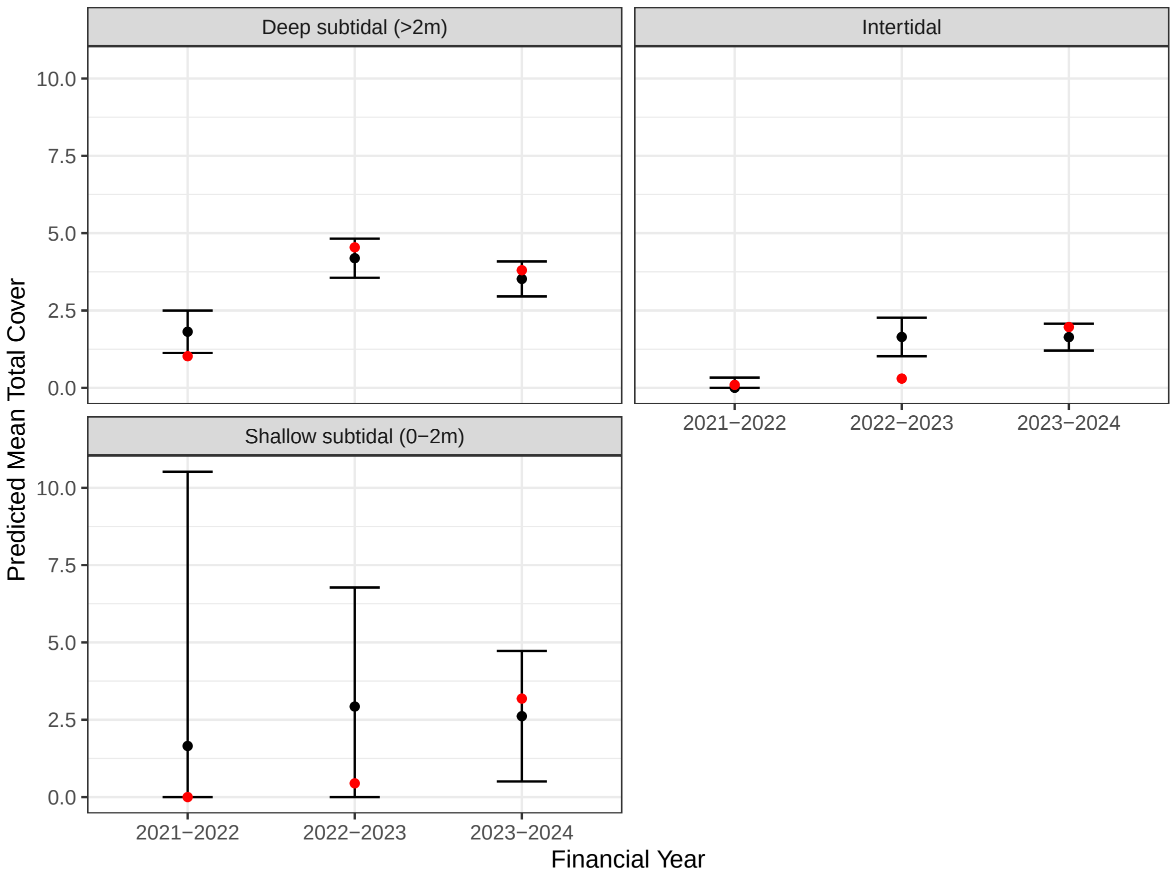

Figure 7: Average total cover predictions by tidal range. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

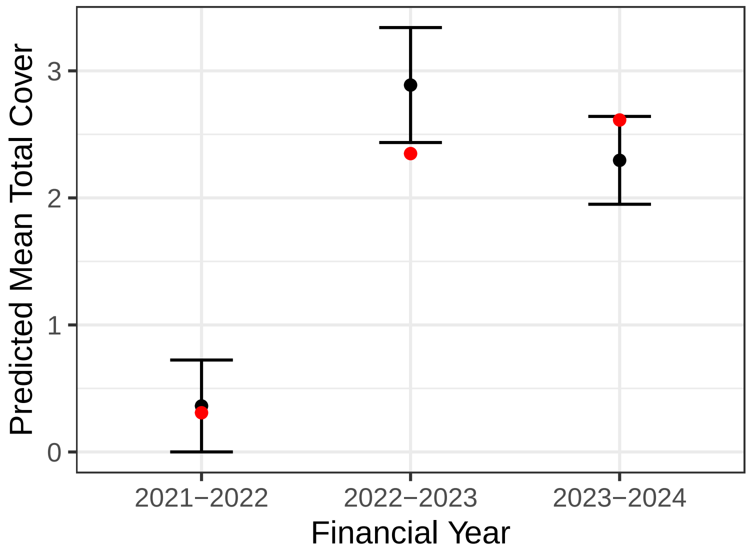

Figure 8: Average total cover predictions in Wide Bay. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

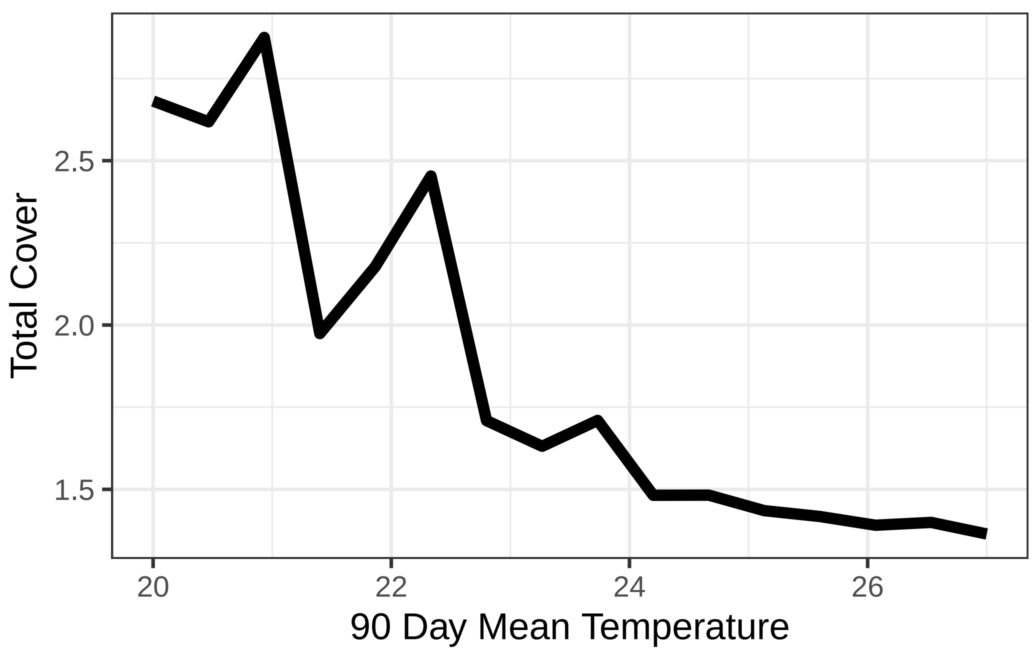

Figure 9: Random forest partial dependence for total cover as a function of 90 day mean temperature

Table 8: Wide Bay out of sample prediction metrics for the various approaches. Mean bias (MBias) tended to be small relative to root-mean-squared-prediction error (RMSPE).

Approach	MBias	RMSPE
RF	-0.03	4.54
SPLM	0.00	4.70
SPGLM-Gamma	1.18	5.12
SPGLM-Beta	-0.16	5.31
LM	0.00	5.45

5 Resilience Modeling

The model described in Section 4 quantifies, on a bay-wide scale, the effects of various water quality and geographic drivers on total seagrass cover. It does not, however, quantity resilience on a site-specific scale. Gilby et al. [2023] provide a framework to quantify resilience on a site-specific scale. They suggest identifying hotspots (i.e., brightspots), sites that are performing better than expected (according to some model fit), as well as coldspots, sites that are performing worse than expected (according to some model fit). They identify hotspots and coldspots based on deviations exceeding expected bounds (based on confidence intervals) based on the model fit. They argue these hotspots and coldspots provide significant utility, both in better understanding the drivers of site-specific resilience as well as informing subsequent management action.

Table 9: Wide Bay resilience variables considered for modeling.

Variable	Effect	Included	Nonlinearity Degree
Coral Habitat Total Area	Fixed	Yes	2
Seagrass Habitat Total Area	Fixed	No	2
Mangrove Habitat Total Area	Fixed	Yes	2
Gastropod Habitat Total Area	Fixed	Yes	NA
Distance to Coral Habitat	Fixed	Yes	3
Distance to Seagrass Habitat	Fixed	Yes	2
Distance to Mangrove Habitat	Fixed	Yes	2
Distance to Gastropod Habitat	Fixed	Yes	2
Species Richness	Fixed	Yes	1
Habitat Count	Fixed	Yes	2
Dominant Species	Fixed	Yes	NA
Bathymetry Mean Depth	Fixed	Yes	2
Bathymetry Mean Curvature	Fixed	No	NA
Bathymetry Mean Slope	Fixed	Yes	2

Table 9: Wide Bay resilience variables considered for modeling.

Variable	Effect	Included	Nonlinearity Degree
EHMP Subzone	Random	Yes	NA

Building from Gilby et al. [2023], we use the residuals from the model in Section 4 as the basis for studying site-specific resilience, linking the values of these residuals to site-specific variables like species habitat, species richness, and bathymetry via a separate spatial linear model, with a random effect for EHMP subzone. These models capture varying nonlinear behavior in the residuals as a function of the resilience variables using B-spline basis functions [de Boor, 1978, Eilers and Marx, 1996] whose degree is informed by random forest partial dependence plots (built from a random forest model on the residuals). While Gilby et al. [2023] focused more on specific hotspots, linking environmental variables to those sites, we used our framework to provide a different set of information – the characterizing of the most important drivers of resilience, on average. Because were were interested in hypotheses not parsimony, we included all potential variables as fixed effects in the model aside from "Gastropod Habitat Total Area" and "Bathymetry Mean Curvature" because there were some challenges processing these data which prohibited their use. These included area of and distance to coral, seagrass, mangrove, and gastropod habitat, species richness and dominant species, habitat count, and bathymetry metrics (Table 9), which have the following structure:

- The "Coral Habitat Total Area" is numeric from 0 msq to 367,906 msq.
- The "Seagrass Habitat Total Area" is numeric from 0 msq to 773,161 msq.
- The "Mangrove Habitat Total Area" is numeric from 0 msq to 241,057 msq.
- The "Distance to Coral Habitat" is numeric from 0 m to 30,563 m.
- The "Distance to Seagrass Habitat" is numeric from 0 m to 7,049 m.
- The "Distance to Mangrove Habitat" is numeric from 0 m to 38,867 m.
- The "Distance to Gastropod Habitat" is numeric from 0 m to 121,609 m.
- The "Species Richness" is numeric (integer) from 0 unique species to 3 unique species.
- The "Habitat Count" is numeric (integer) from 0 unique habitats to 3 unique habitats.
- The "Dominant Species" has five levels: "Absent", "HD", "HO", "HS", and "ZMHU".
- The "Bathymetry Mean Depth" is numeric from -30.58 m to 0.25 m.
- The "Bathymetry Mean Slope" is numeric from 0.004 degrees to 3.93 degrees.
- The "Zone Class" variable has six levels: Central, East, Great Sandy Strait, Hervey Bay, Southern, Southwest (Figure 3).

5.1 Model Fit

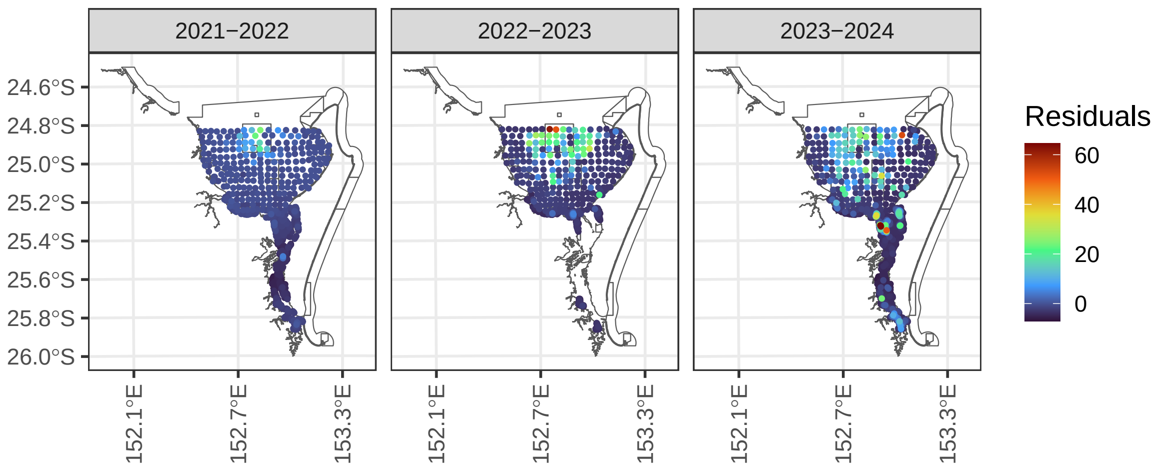

Figure 10: Residuals for reslience analysis in Wide Bay by financial year.

Table 10: ANOVA table of covariates in the Moreton Bay resilience model.

Effects	df	Chi-sq	p.value
Species Richness	1	238.769	0.000
Dominant Species	4	68.448	0.000
Distance to Coral Habitat	2	51.856	0.000
Distance to Gastropod Habitat	2	3.196	0.202
Distance to Mangrove Habitat	2	2.913	0.233
Seagrass Habitat Total Area	2	2.834	0.242
Mangrove Habitat Total Area	2	1.161	0.560
Bathymetry Mean Slope	2	0.699	0.705
Distance to Seagrass Habitat	2	0.551	0.759
Bathymetry Mean Depth	2	0.435	0.804
Coral Habitat Total Area	2	0.094	0.954
Habitat Count	2	0.042	0.979

From Table 10, we find strong evidence (p-value < 0.001) that species richness, dominant species, and distance to coral habitat are associated with resilience:

- Species Richness: An increase of species richness by one species was associated with an average increase in resilience by approximately five points (Figure 11).
- Dominant Species: The estimated (marginal) average resilience score is largest for HS (-0.17), followed by HD (-2.9), HO (-3.2), and ZMHU (-3.9). The overall ranking is most important, as the negative average scores are in part likely due to the heavy skew in the data.
- Distance from Coral Habitat: From zero to around 10,000 m from coral habitat, there is a decreasing trend with resilience (i.e., increasing distance decreases resilience). Upwards of 10,000 m, there is an increasing trend with resilience (i.e., increasing distance increases resilience).

We found little evidence (p-value > 0.1) the remaining variables were associated with resilience (after controlling for other modeled variables).

5.2 Alternative Approaches

Another way we studied resilience was using a random forest (fit to the same residuals). Largely, we found similar results as the spatial linear model. Notably, for the random forest:

- Species richness was most important, with a general increasing trend in resilience with richness.
- Distance from mangrove habitat was second-most important, with a general increasing trend in resilience with distance from mangroves

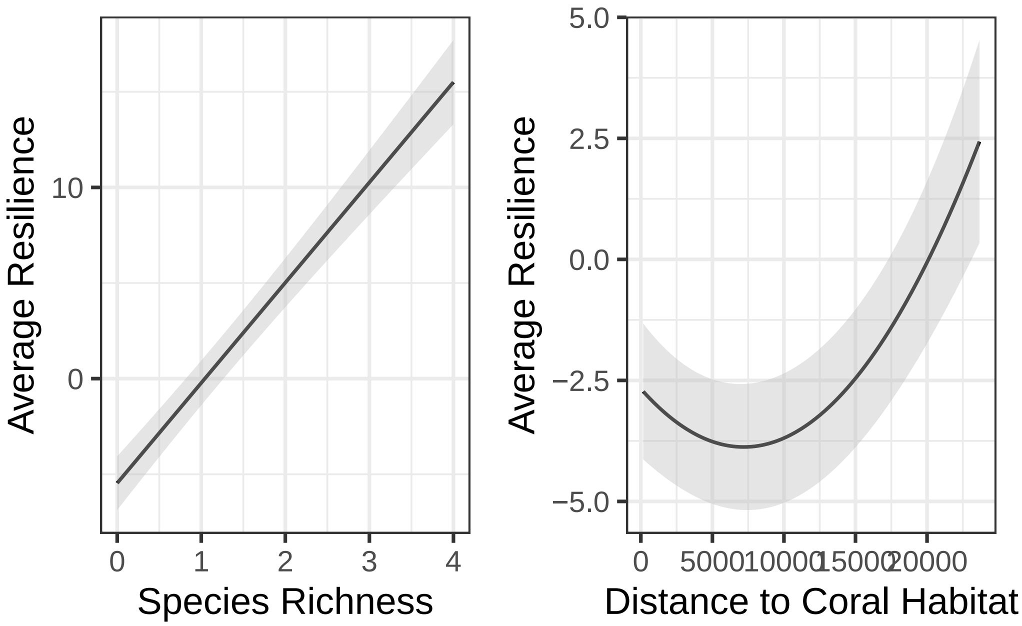

Figure 11: Average species richness (left) and distance to coral habitat (right) effects on resilience. Predictions (and confidence intervals) were evaluated at the means of all other numeric variables and for the ZHMU species in the Southern Banks (similar trends occur for both variables among the remaining species and subzones).

- Dominant species was third-most important, with HS being the most resilient followed by HO, ZMHU, and HD.
- Bathymetry mean depth was fourth-most important, with a general increasing trend in resilience with depth.
- Distance from coral habitat was fifth-most important, with a similar trend quadratic trend (**?**) as in the spatial linear model.
- The remaining variables (gastropod distance, seagrass area, bathymetry slope, seagrass distance, zone class, habitat count, mangrove area, and coral area) were least important.

Other alternatives to explore involve using the residuals from water quality and geographic models fit using the spatial generalized linear models and random forests. These residuals could be analyzed using spatial linear models (recall that the residuals are unconstrained) and random forests. We expect these alternative approaches to yield generally similar results, but there could be nontrivial differences given the heavy skew of the data, however.

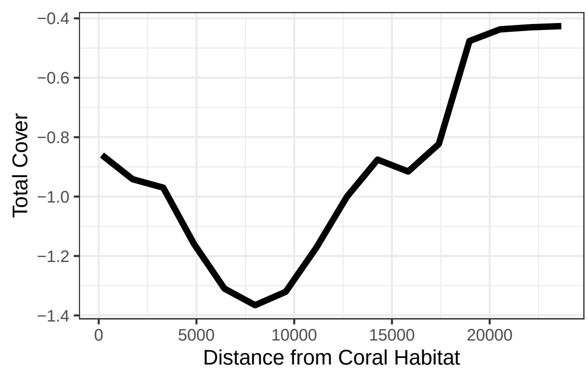

Figure 12: Random forest partial dependence for total cover as a function of distance from coral habitat.

6 Limitations and Future Suggestions

While the spatial linear model was useful for characterizing broad patterns related to tidal range, sediment type, temperature, and turbidity and site-specific patterns capturing resilience, all while accounting for spatial dependence, there are some limitations. First, the eReefs water quality data are modeled, not directly observed in the field, potentially propagating modeling error. Second, there are many correlated water quality variables, making it hard (for any model) to isolate the effects of each component of water quality pre- and post-flood. Third, there is a lack of baseline mean salinity in the 90 days prior to sampling, which makes it challenging to directly study the effect of mean salinity during the flood period. Fourth, there is a lack of data immediately after each flood – nearly all of the data are after both floods, making it challenging to isolate the contribution from each flood separately. Fifth, there was no pre-flood data, making it challenging to compare against a pre-flood baseline. Sixth, there is a lack of addition data sources that could help quantify the impact of various explanatory variables on total pre and post-flood, potentially varying across both space and time (e.g., light attenuation, biological interactions, etc.). Seventh, the locations at which data were collected were not randomly sampled. Randomly sampling sites helps to guard against sampling preferentially [Diggle et al., 2010] and provides more robust and representative data sources [Dumelle et al., 2022]. Eight, the data have heavy skew and many observations without seagrass. While we explored several approaches and ultimately preferred the spatial linear model because of its interpretability and utility with block Kriging, the corresponding assumptions are less reliable than if the data had a more dense set of nonzero seagrass cover observations. No matter the approch ultimately used to analyze the data, highly skewed data with many zeros complicate modeling.

7 Revisiting the Research Questions

Research Question 1: Can we identify water quality covariates that act as triggers for seagrass cover management responses? Are these triggers different based on whether they result from a specific event (e.g., flood) or are ambient?

Research Question 2: Did seagrass cover improve after the flood event(s)? If so, how and where?

Research Question 3: Can we identify site-specific resilience indicators, which describe where recovery is likely to be enhanced or impeded? What factors contribute to seagrass resilience?

8 Discussion

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